

Component Guided Wire Extraction From Hand Drawn Circuit Schematics Using Adaptive Threshold Fusion and Topology Aware Skeleton Graphs

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Abstract

Digitization of hand drawn circuit schematics remains difficult because thin wire traces are easily corrupted by paper texture, scanner artifacts, uneven lighting, and component overlap. A component guided hybrid method is presented in which oriented component detections are used for occlusion and region of interest definition, while wire extraction is performed with a deterministic computer vision pipeline. In the revised method, adaptive threshold masks are converted into skeleton graphs, a conservative Sauvola branch is supplemented by a gated adaptive Gaussian recovery branch, candidate paths are scored by geometric support and component proximity, and a topology aware selection stage is used to suppress redundant alternatives. In the final variant, recovery candidates are further constrained by anchor requirements, class aware port cues, and parallel duplicate rejection. No wire specific training is required for the main method. On a benchmark of 23 hand drawn circuit images with 300 annotated wire segments, the reproduced reference pipeline achieved a global F1 score of 0.7066. The earlier topology guided skeleton graph method achieved an F1 score of 0.7830. The final hybrid recovery method achieved an F1 score of 0.8215, with precision

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0.7275 and recall 0.9433. The method is therefore shown to improve both reproducibility and wire extraction accuracy while preserving a transparent pipeline that can be inspected stage by stage.

Keywords: Circuit diagram digitization, Wire extraction, Skeleton graph, Adaptive thresholding, Document image analysis, Hybrid computer vision

1. Specifications table

Table 1: Specifications table

Subject area	Computer Science
More specific subject area	Computer vision, document analysis, circuit diagram digitization
Name of the method	Component Guided Wire Extraction With Adaptive Threshold Fusion and Topology Aware Skeleton Graphs
Name and reference of original method	Sauvola adaptive document image binarization [2] and a reproduced repository baseline for circuit wire extraction
Resource availability	Pipeline repository: https://github.com/boscochanam/circuit-digitization . Component dataset: CGHD1152 and HDC subset documented in the repository. Wire benchmark: 23 annotated images with 300 wire segments, provided by the authors.
Main validation metric	Point based wire segment F1 score with a 20 px endpoint distance tolerance; best result 0.8215 on the 23 image benchmark

2. Background

Digitization of hand drawn circuit schematics has remained challenging because reliable interpretation depends on both symbol localization and recovery of thin connecting wires. Recent progress has been reported in handwritten and engineering diagram recognition through detector based pipelines, graph enhanced postprocessing, and domain specific datasets [1,

12, 21, 13, 14, 15, 16, 17, 18]. In most of these studies, strong performance has been obtained for component or symbol detection, whereas wire or pipeline recovery has remained sensitive to local clutter, rotation, scan noise, and low contrast traces. The resulting gap has limited the direct conversion of offline sketches into simulation ready circuit structure.

The document analysis literature has shown that binarization and line recovery remain decisive when fine strokes must be preserved. Classical thresholding methods such as Otsu, Niblack, and Sauvola remain widely used because they are interpretable and easy to control [3, 4, 2]. Their limitations on degraded documents have been studied extensively, and improved adaptive strategies have been proposed through contrast based thresholding, local background estimation, and efficient integral image implementations [5, 6, 7]. At the same time, skeletonization and line extraction have been developed into mature tools for preserving topology and tracing elongated structures [8, 9, 10, 11]. These ideas are especially relevant when the objective is not merely foreground segmentation, but recovery of sparse geometric structure from weak and fragmented strokes.

Related work in engineering drawings has also indicated that symbol recognition alone is insufficient. Deep detectors have been used successfully for symbols in piping and instrumentation diagrams, single line electrical drawings, and related symbolic sketches, yet downstream structural inference still requires domain aware processing of lines, ports, and connectivity [14, 15, 16, 17, 18]. Hand drawn circuit studies have similarly relied on object detection followed by node or line reasoning, often through Hough based or heuristic geometric procedures [12]. These observations suggest that hybrid methods remain attractive when wire specific labeled data are limited.

In the earlier manuscript version, a hybrid strategy was adopted in which component detections were used as priors while wire extraction was performed by a classical pipeline based on Sauvola thresholding, connected component analysis, and segment merging. That formulation established the value of component guided preprocessing, but the strongest validated result from the current repository differs materially from the earlier draft. The merge based wire result of 0.647 no longer reflects the best reproducible method. A reproduced reference pipeline now yields an F1 score of 0.7066, and the strongest validated method in the present work yields an F1 score of 0.8215 on the same 23 image benchmark.

The present revision has therefore been focused on the wire method itself. The key observation was that line fitting directly from connected compo-

nents compresses image structure too early. Fragmented traces, redundant segments, and ambiguous endpoints were repeatedly produced in dense regions. To address this issue, a skeleton graph representation was introduced. Adaptive threshold masks were converted into graph paths, candidate wire hypotheses were scored by structural and component guided cues, and a topology aware selection step was applied to preserve a sparse and plausible explanation of the circuit layout.

The resulting manuscript has been written as a MethodsX article because the principal contribution is methodological. A reproducible hybrid wire extraction procedure is documented, its behavior is validated on a manually annotated benchmark, and its design choices are presented in a form that can be inspected and reused in related document analysis tasks.

3. Method details

The method is organized as a component guided wire extraction pipeline. Oriented component labels are used to suppress non wire structures before thresholding, while the final wire decision is made by a deterministic graph based postprocessing stage. The benchmark implementation is available in the repository and was validated against the reproduced reference pipeline.

Datasets and benchmark definition

Two data sources were used. Component priors were obtained from the CGHD1152 handwritten circuit corpus and its HDC recognition subset, which provide oriented component annotations across 58 classes [1]. Wire extraction was evaluated on 23 hand drawn schematic images at 704 by 704 px resolution with 300 manually annotated wire segments in YOLO OBB format. Each quadrilateral annotation was converted into a centerline segment through the short edge midpoint procedure used by the benchmark evaluator.

Evaluation was performed with the repository metric in which a detected segment is counted as a true positive when its endpoints lie within a mean 20 px tolerance of the nearest ground truth segment. Precision, recall, F1 score, and redundant detections were reported. During the later hybrid development stage, a secondary synthetic sanity check was also maintained by running each candidate configuration on procedurally generated wire images

from the repository synthetic data generator. Because these synthetic images do not provide external component priors, they were used as a stability check rather than as the primary model selection signal.

Component guided preprocessing

For each benchmark image, the corresponding HDC component label file was matched by pixel comparison across augmentation variants. This step was required because multiple transformed copies of the same base image were present in the component dataset. Once the correct label file had been identified, component polygons and text polygons were parsed and used in two preprocessing operations.

First, each polygon was occluded with the local median intensity estimated from a margin around the polygon. This operation suppressed symbol strokes and printed text that would otherwise be converted into false wire evidence. Second, the image was cropped to the union of all component boxes with 10 px padding. This operation reduced border clutter and focused subsequent processing on the circuit region.

Adaptive threshold mask generation

Wire evidence was generated from Sauvola thresholding [2]. This choice was retained because local adaptive thresholding has repeatedly been found to outperform global thresholding on degraded document content and thin foreground strokes [3, 4, 5, 6, 7]. The local threshold was computed in a 67 by 67 window with primary parameter $k = 0.285$. A secondary Sauvola mask at $k = 0.25$ was also generated. In the final hybrid method, these masks were treated as conservative evidence sources, while a gated adaptive Gaussian recovery branch was introduced to recover weak traces that were missed by the primary branch.

This stage was designed to preserve faint traces while avoiding excessive sensitivity to low frequency paper intensity variation. In contrast with the earlier merge based pipeline, no global line merging stage was introduced at this point. Instead, the permissive branch was allowed to contribute only when candidate paths were supported by component proximity and graph consistency tests.

Additional threshold families were also evaluated within the same benchmark harness. Global Otsu thresholding, triangle thresholding, adaptive mean thresholding, adaptive Gaussian thresholding, and simple fusion variants were substituted into the same downstream extraction and evaluation

procedure. This comparison was introduced to test whether the strongest result depended specifically on Sauvola thresholding or on later graph based reasoning stages.

Skeleton graph extraction

The fused binary foreground mask was converted into a one pixel skeleton. Pixels with degree other than two were treated as graph nodes, and skeleton chains between such nodes were traced as candidate paths. Each path was converted into a line hypothesis through its terminal coordinates. Very short paths were removed through a minimum graph path length threshold. This formulation was informed by classic thinning and skeletonization work, in which centerline preservation and topological stability are treated as the key requirements [8, 9].

This representation was preferred over direct connected component line fitting because it retained local topology for a longer portion of the pipeline. Junctions, branch terminations, and elongated traces were therefore represented more faithfully than in the earlier contour extremum formulation. It also avoided premature commitment to a single contour extremum pair, which is a known weakness when line evidence is fragmented or when several near collinear alternatives coexist.

Candidate scoring and selection

Each candidate path was assigned a score from four sources of evidence:

1. geometric length,
2. average support across threshold masks,
3. proximity of endpoints to component boundaries,
4. proximity of endpoints to plausible component side midpoints that act as coarse port surrogates.

The scored candidates were then filtered by a topology aware selection stage. A sparse set of non overlapping paths was retained, while weak alternatives that did not add new structural support were removed. This step differed from the earlier deduplication strategy in two important ways. First, candidates were ranked rather than treated symmetrically. Second, acceptance was conditioned on how well a candidate extended the current graph explanation of the circuit.

After topology selection, score cluster deduplication was applied to suppress near collinear duplicates. The role of this step was similar in spirit to

line selection methods based on Hough style voting or direct line segment detection, but selection was performed over graph paths rather than over edge pixels alone [10, 11]. In the final hybrid configuration, a secondary adaptive Gaussian branch was executed after the primary topology guided Sauvola pass. Its candidates were admitted only when they remained near components, satisfied explicit anchor requirements, linked coherently to unresolved graph endpoints, and did not create near parallel duplicates around accepted paths. In the last revision stage, the anchor test itself was tightened by using class aware port surrogates derived from oriented component polygons rather than from generic bounding box edges. Two terminal components were therefore encouraged to connect along their short side midpoints, whereas non connective classes such as text and crossover labels were removed from the anchor pool. An additional endpoint driven repair variant was also examined in which recovery candidates were seeded only from dangling endpoints of the conservative topology graph. That variant reduced false positives more aggressively than the permissive hybrid branch, but a larger recall loss was incurred and the overall benchmark score was not improved. The final main method is therefore referred to in this paper as the hybrid recovery method.

Method summary

The full revised architecture is shown in Fig. 1. The original connected component branch remained useful as a control, but the strongest performance was obtained when graph extraction and topology aware selection were introduced.

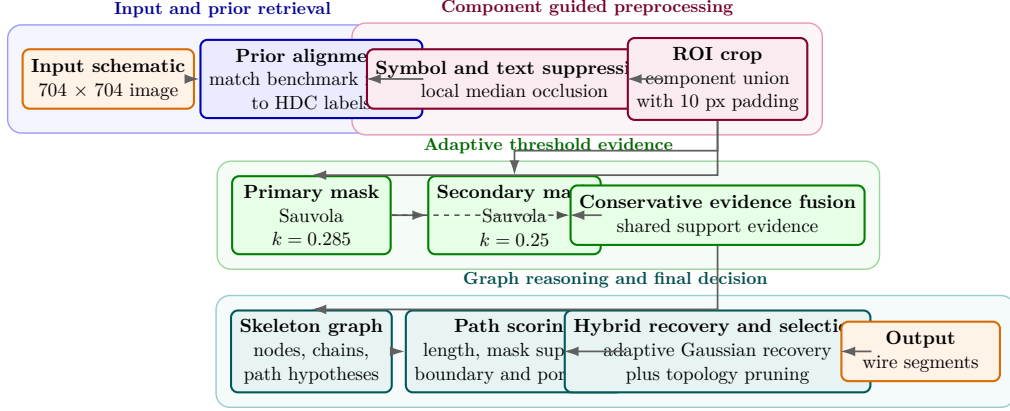


Figure 1: Revised wire extraction architecture. Component priors are first aligned to each benchmark image and used to suppress symbols, text, and border clutter. Conservative Sauvola evidence is fused into a shared support representation, from which a skeleton graph is extracted. Adaptive Gaussian recovery is then used selectively for weak paths before final topology aware selection is applied.

4. Method validation

The reproduced reference implementation yielded an F1 score of 0.7066 on the 23 image benchmark. When the strongest refined classical pipeline was used, the F1 score increased to 0.7492. The topology guided skeleton graph method increased the benchmark score further to 0.7830. The final hybrid recovery method increased the score again to 0.8215 by supplementing the conservative Sauvola branch with a tightly gated adaptive Gaussian recovery branch and class aware anchor cues.

Table 2: Benchmark comparison on 23 hand drawn circuit schematic images with 300 annotated wire segments.

Method	F1	Precision	Recall	TP	FP	FN	Red
Reference baseline	0.7066	0.6169	0.8267	248	70	52	84
Refined classical pipeline	0.7492	0.7236	0.7767	233	43	67	46
Topology guided skeleton graph	0.7830	0.7411	0.8300	249	48	51	39
Hybrid recovery method	0.8215	0.7275	0.9433	283	63	17	43

The main improvement was not obtained through a single threshold change. Instead, the increase was produced by a sequence of architectural

changes. Once connected component fitting was replaced by skeleton graph extraction, and once topology aware selection was introduced, redundant alternatives were removed more effectively while the strongest supported paths were preserved. The final hybrid recovery method then recovered additional weak paths while constraining false positives through component aware gating, explicit anchor requirements, unresolved endpoint linkage, class aware port surrogates, and parallel duplicate rejection. Relative to the reproduced baseline, true positives increased from 248 to 283, false positives decreased from 70 to 63, and the final F1 score rose from 0.7066 to 0.8215.

The contrast between the topology guided and hybrid variants was also informative. The topology guided skeleton graph method retained the highest precision among the leading methods, whereas the hybrid recovery method produced a much larger recall gain. The best overall F1 score was obtained only after the adaptive Gaussian branch was constrained tightly enough to behave as a selective recovery mechanism rather than as a global replacement. In the final variant, recovery candidates were filtered with class aware port cues that removed several structurally implausible attachments, which reduced false positives from 71 to 63 while maintaining most of the recall gain from the earlier hybrid stage.

An additional endpoint driven recovery experiment was then performed in which permissive candidates were seeded only from unresolved endpoints in the conservative topology graph and were accepted only when they repaired short component aware gaps. This modification reduced false positives further relative to the permissive hybrid family, but the best such variant reached an F1 score of 0.7957 because too much recall was surrendered. The result suggested that the conservative topology graph is an effective precision prior, but that a stricter repair only formulation is not sufficient on its own to replace the final hybrid gate.

Thresholding comparison

Thresholding was then varied while the same benchmark harness, pre-processing, extraction, and evaluation logic were preserved. The objective was to determine whether a different binarization family could improve the present method or reveal a more favorable precision recall tradeoff.

Table 3: Thresholding comparison on the 23 image wire benchmark.

Threshold setting	F1	Precision	Recall	TP	FP	FN	Red
Reference Sauvola baseline	0.7066	0.6169	0.8267	248	70	52	84
Topology guided Sauvola method	0.7830	0.7411	0.8300	249	48	51	39
Final hybrid recovery method	0.8215	0.7275	0.9433	283	63	17	43
Adaptive Gaussian skeleton method	0.7487	0.6164	0.9533	286	126	14	52
Tuned Adaptive Gaussian skeleton method	0.7576	0.6286	0.9533	286	117	14	52
Adaptive mean skeleton method	0.7355	0.6000	0.9500	285	120	15	70
Otsu skeleton method	0.5671	0.4571	0.7467	224	207	76	59
Sauvola and Otsu fusion	0.5915	0.4648	0.8133	244	210	56	71

The comparison showed that global Otsu thresholding was not competitive for this dataset. Both the pure Otsu skeleton method and the Sauvola and Otsu fusion setting produced large false positive counts and substantially lower F1 scores than the final method. Adaptive mean and adaptive Gaussian thresholding were more promising because much higher recall was obtained. The strongest tuned adaptive Gaussian setting reached an F1 score of 0.7576 with recall 0.9533, but this gain in recall was offset by a large increase in false positives. The thresholding study therefore supported two conclusions. First, the strongest global result still depended on Sauvola based local thresholding as the conservative primary branch. Second, adaptive Gaussian thresholding became useful only when it was constrained as a selective recovery branch inside the final hybrid method.

The same late stage variants were also checked on synthetic images from the repository generator. Those synthetic runs did not separate the leading component guided variants in a meaningful way because the generated images were evaluated without external component priors. Synthetic data were therefore retained as a secondary stability check, while the real 23 image benchmark remained the decisive model selection setting.

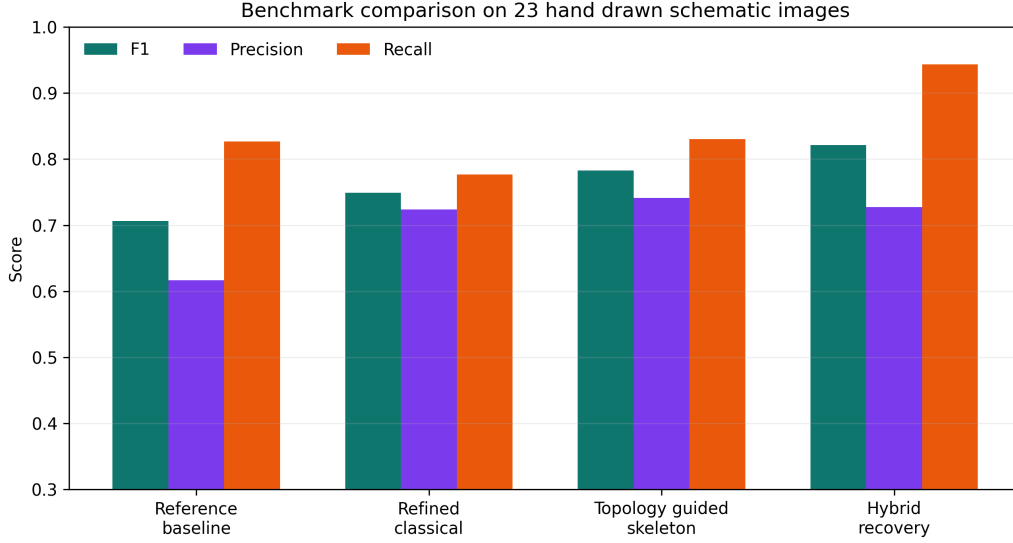


Figure 2: Aggregate benchmark comparison for the reproduced reference baseline, the strongest earlier classical branch, the topology guided skeleton graph method, and the final hybrid recovery method. The hybrid formulation was found to provide the highest F1 score through a large recall gain while preserving substantially better precision than an unconstrained permissive threshold branch.

Qualitative behavior is shown in Fig. 3. A perfect recovery case is included to demonstrate the expected output on clean circuit structure. A difficult recall case is included to show that weak or fragmented traces are still the dominant failure mode. A redundancy heavy case is also included because densely connected layouts still induce extra line hypotheses even after topology filtering. For each example, intermediate stages are shown so that the effect of occlusion, conservative thresholding, permissive recovery, and final selection can be inspected directly.

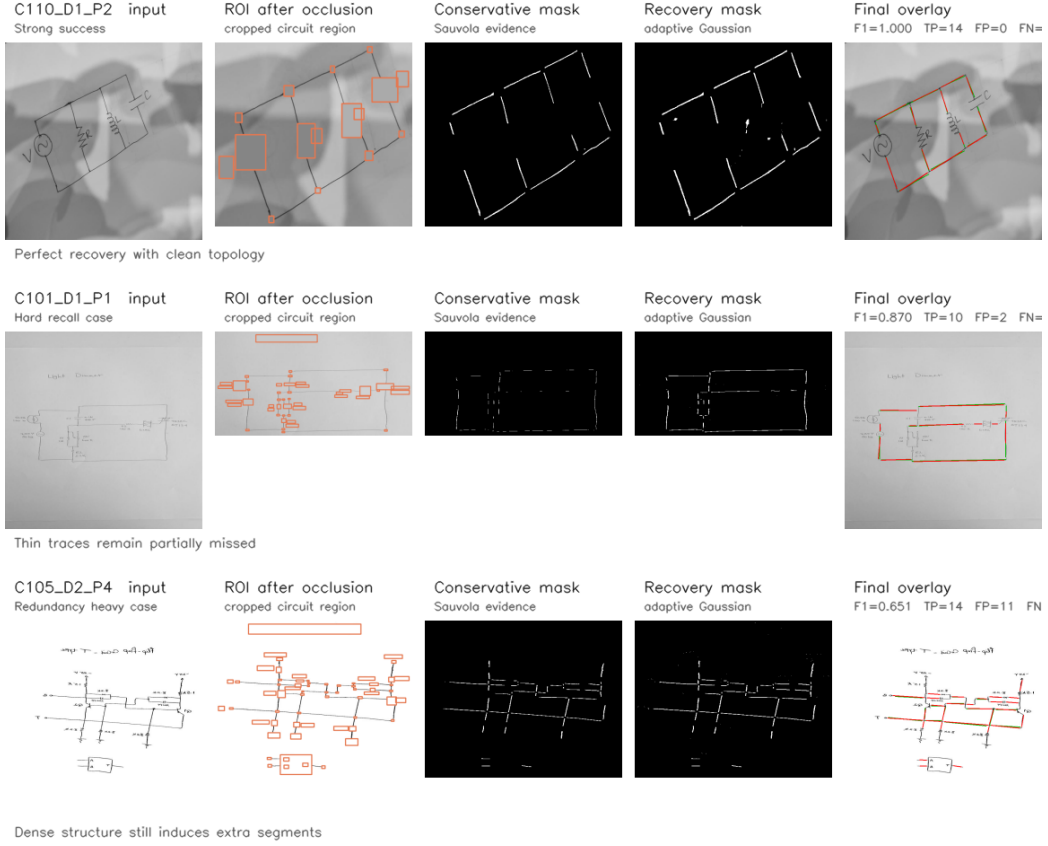


Figure 3: Representative pipeline stages from the final hybrid recovery method. For each example, the input image, cropped region after component and text suppression, conservative Sauvola mask, adaptive Gaussian recovery mask, and final prediction overlay are shown. Green segments denote ground truth and red segments denote predicted wire segments.

The present benchmark was also used to compare two conclusions from the earlier draft. First, the old merge based result of 0.647 was no longer supported once the corrected repository benchmark was reproduced. Second, the best current method no longer depends on collinear merging or length filtering. The strongest reproducible result was instead obtained when topology aware pruning was combined with a tightly gated recovery branch after graph extraction.

5. Limitations

Several limitations should still be acknowledged.

First, the method remains dependent on accurate component priors. Incorrect component labels or mismatched augmentation variants can degrade both occlusion and region of interest selection. The revised benchmark scripts now reproduce the intended pairing procedure, but this dependency remains part of the method.

Second, the hardest remaining failures are still recall driven. Very faint strokes and fragmented traces near crowded components are not always preserved, even after threshold fusion and graph selection. This behavior was observed in images such as C101_D1_P1 and C10_D2_P3.

Third, redundant alternatives can still be produced in dense local structure. The topology aware selection stage reduced this effect substantially, but dense cases such as C105_D2_P4 and C102_D1_P2 still produced extra segments.

Fourth, the benchmark is small. Although the 23 image set is carefully annotated and was sufficient for controlled method comparison, broader generalization across additional drawing styles and acquisition conditions should still be validated.

Finally, the present synthetic generator remained useful for sanity checking, but it was less decisive for the later component guided hybrids because the generated images were not paired with external component priors in the same manner as the real benchmark. A quick learned wire branch was also explored during the present revision, but it did not surpass the deterministic graph based method under the available training setup. The current manuscript is therefore focused on the strongest validated hybrid method rather than on a partially trained learned alternative.

6. CRediT author statement

Bosco Chanam: Conceptualization, methodology, supervision, writing review and editing.

Chris Dcosta: Data curation, software, validation, writing review and editing.

Pranavesh Kumar Talupuri: Methodology, software, formal analysis, visualization, writing of the revised manuscript.

7. Data availability

The pipeline repository is available at <https://github.com/boscochanam/circuit-digitization>. The component dataset source is the public CGHD1152 corpus [1], together with the HDC recognition subset documented in the repository. The 23 image wire benchmark with 300 annotated wire segments is maintained by the authors. Benchmark outputs, figure assets, and manuscript source files are contained in the present project workspace.

8. Acknowledgments

Institutional support from the Symbiosis Centre for Applied Artificial Intelligence and the University of Southern California is gratefully acknowledged.

9. Declaration of interests

The authors declare that no known competing financial interests or personal relationships could have appeared to influence the work reported in this paper.

Appendix A. Per image benchmark results

The per image results for the final hybrid recovery benchmark run are listed in Table A.4. These values were generated directly from the repository benchmark output and were not entered manually.

Table A.4: Per image benchmark results for the final hybrid recovery method.

Image	F1	TP	FP	FN	Red	Tags
C100_D1_P1	0.889	24	1	3	2	–
C100_D1_P3	0.704	19	7	5	4	fp_heavy
C100_D2_P4	0.810	17	3	0	5	redundancy_heavy
C101_D1_P1	0.870	10	2	0	1	–
C102_D1_P2	0.485	8	11	0	6	fp_heavy, redundancy_heavy
C103_D2_P1	0.737	7	1	0	4	redundancy_heavy
C103_D2_P4	0.941	8	0	0	1	–
C104_D1_P3	0.615	8	5	0	5	fp_heavy, redundancy_heavy
C104_D1_P4	0.857	9	1	0	2	–
C104_D2_P2	1.000	13	0	0	0	–
C104_D2_P3	1.000	13	0	0	0	–
C104_D2_P4	0.917	11	2	0	0	–
C105_D1_P1	0.800	16	6	1	1	fp_heavy
C105_D1_P4	0.681	16	9	0	6	fp_heavy, redundancy_heavy
C105_D2_P4	0.651	14	11	0	4	fp_heavy, redundancy_heavy
C109_D2_P3	1.000	7	0	0	0	–
C10_D2_P3	0.714	5	1	3	0	–
C110_D1_P2	1.000	14	0	0	0	–
C111_D1_P1	0.957	11	1	0	0	–
C112_D1_P1	0.870	20	0	5	1	–
C113_D2_P3	1.000	7	0	0	0	–
C115_D2_P3	1.000	7	0	0	0	–
C127_D2_P1	0.927	19	2	0	1	–

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